CSE 486/586 Group Project

Project Description: Human Pose Detection using MMVR Dataset

Introduction

Human pose detection is a fundamental task in computer vision and artificial intelligence, enabling systems to estimate and track human body keypoints in real time. Applications range from activity recognition and sports analytics to human-computer interaction and autonomous systems that require an understanding of human behavior in dynamic environments. Traditional vision-based pose estimation techniques often rely solely on RGB imagery, which can struggle in low-light conditions, occlusions, or environments with visual clutter.

To address these challenges, multi-modal sensing approaches have emerged. The MMVR (Multi-Modal Vision and Radar) dataset integrates high-resolution RGB video with mmWave radar signals, providing complementary visual and motion cues. While RGB data captures rich spatial information, radar sensing contributes robust motion and range data, especially under adverse conditions such as poor lighting or partial occlusions.

This project aims to develop AI-based methods for pose detection leveraging the MMVR dataset. Students will explore deep learning architectures for 2D and 3D human pose estimation, integrate radar features to enhance robustness, and evaluate system performance across diverse scenarios.

The challenge lies in developing robust human pose detection methods that overcome limitations of vision-only approaches, such as poor lighting and occlusions. This project addresses the problem by:

1. Detecting 2D and 3D human body keypoints using mmWave data with advanced deep learning models.
2. Integrating mmWave radar signals to capture complementary motion and range information, enhancing pose estimation accuracy in challenging environments.
3. Leveraging the multi-modal MMVR dataset to train and evaluate fused models that improve robustness and reliability across diverse real-world conditions.

Dataset: MMVR

The Millimeter-wave Multi-View Radar (MMVR) dataset is a large-scale indoor radar perception dataset designed to advance research in multi-modal sensing and human perception. It contains 345,000 multi-view high-resolution radar heatmap frames collected from 25 human subjects

across 6 diverse rooms over 9 separate days. The dataset is richly annotated with approximately 446,000 bounding boxes and segmentation instances, along with 7.59 million human keypoints, supporting three major perception tasks: object detection, pose estimation, and instance segmentation.
Dataset Link: [Millimeter-wave Multi-View mRadar (MMVR) Dataset]

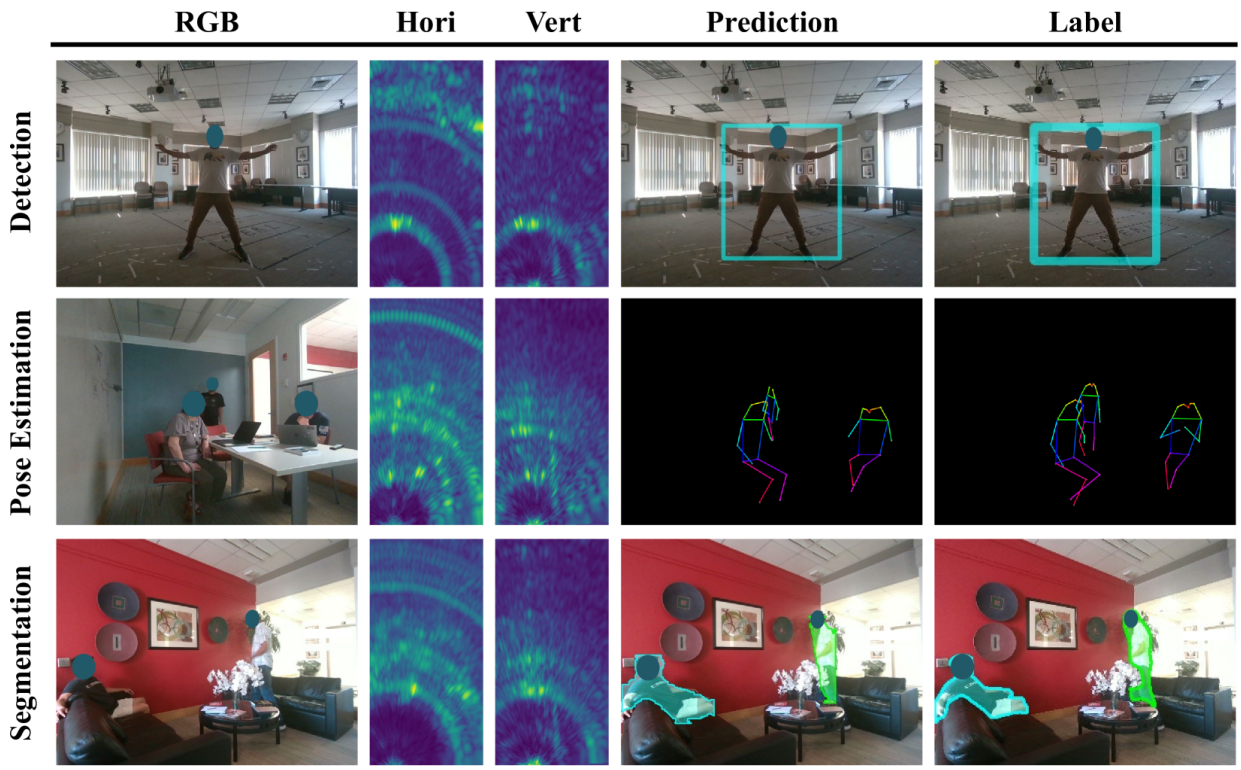

Fig: Example from dataset with sensor data
(https://dl.acm.org/doi/10.1007/978-3-031-72986-7_18)

Objectives

1. **Pose Detection (RGB-based):**
    a. Utilize state-of-the-art deep learning models (e.g., HRNet, OpenPose, or ViTPose, or your own model) to detect key human body joints from RGB video frames.
    b. Radar-Enhanced Pose Estimation:
        i. Fuse mmWave radar data with visual features to improve accuracy in challenging scenarios (e.g., low light, occlusion, motion blur).
        ii. 3D Pose Reconstruction:
2. **Performance Evaluation:** Keypoint Accuracy: Measure using PCK (Percentage of Correct Keypoints) and OKS (Object Keypoint Similarity).

a. Robustness Analysis**:** Evaluate performance under adverse conditions (occlusion, darkness, clutter).
b. Generalization: Assess model transferability to unseen environments.
c. Visualize a different validation matrix for validation

Groups Description

To better facilitate the learning objectives, a Group Project will be included in this course. Since you have already formed your groups, please follow the instructions below to complete the project.

- Undergraduates are recommended to allocate the workload so that they can do it in parallel.
- Graduate students should work together as a group to complete the task.

Note: The way you develop your system is open-ended. You are encouraged to research more into potential methods you can use to complete the project.

What to submit (Due by 04/07/2026)

Final Submission Files (ZIP File + Two Reports)

(1) ZIP File

(Upload in a ZIP - name it as **UniqueID_GroupProj_1.zip** - Change the filename according to your group members' UniqueID. For example, **yuh3**_GroupProj_1.zip, where **yuh3** is the group member's UniqueID. If your group has more than one member, you may include just one group member's UniqueID in the filename. I will then look up the other group members based on that UniqueID and the email you sent me earlier).

Python Code (`.ipynb/.py`)

- Well-documented code for:
  - Data processing
  - Models
  - Visualization of results
  - Evaluation

- Code should be commented and clean.

Your Trained Model (`.pth`)

- Your final model after the training process.

(2) Two Reports

The project will include **two reports**: one is a 3-page technical report, excluding references (if any), and the other is a 1-page team contribution report. Please see the details below:

- For the technical report, **please include *the names of all group members* in the report**. The report should also include a description of your work, the model architecture, the dataset, the results (with figures), and any relevant equations.
- The team contribution report will describe each individual's contributions. The report should include:
    - A breakdown of tasks assigned to each member.
    - A brief description of what each member completed.
- Templates for the reports will be provided on Canvas under the **Project** folder.
- To reduce the workload, there will be NO presentation for this project.

Grading Rubric – AI Class Group Project

Category	**Exemplary**	**Proficient**	**Developing**	**Needs Improvement**	**Possible Points**
Data Preprocessing	Complete and well-documented data preprocessing pipeline. Dataset fully synchronized, calibrated, and prepared for training. Clear explanation of preprocessing steps and data quality validation. Model selection is justified with a training plan outlined.	Data preprocessing is mostly complete with minor gaps or missing documentation. Model ready for training with basic rationale provided.	Partial data preprocessing; some critical calibration or synchronization steps are missing. Model readiness is unclear or incomplete.	Minimal or no data preprocessing done; data not ready for training. No model preparation evident.	20

Model Selection & Training Complete	Model fully trained with thorough documentation of training setup, hyperparameters, and performance on training/validation data. Clear explanation of model architecture and reasoning behind selection.	Model training is mostly complete with basic documentation. Some details on model choices and training metrics.	Model partially trained; weak documentation or missing key training parameters.	No meaningful model training or documentation provided.	25
Additional Models & Distance Estimation Complete	Implementation and training of one additional model completed with comparative analysis. Distance estimation using mmWave radar has been successfully integrated and validated. Clear presentation of intermediate results.	One additional model was implemented with basic results. Distance estimation is mostly complete with minor gaps.	One additional model was attempted, but was partially done. Limited analysis.	No attempts at additional models or distance estimation.	25
Final Report & Code (Visualization & Submission)	Comprehensive visualization of all models' detection, distance, and localization results. Clear, well-structured final report and clean, well-documented code submitted on time. Strong reflection on outcomes and challenges.	Good visualization and report submitted with mostly complete code. Some minor clarity or documentation issues.	Partial visualization or incomplete final report/code submission. Reflective analysis is weak or missing.	Final report or code missing or largely incomplete. Poor presentation of results.	30
Total Score					100

Still have questions?

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